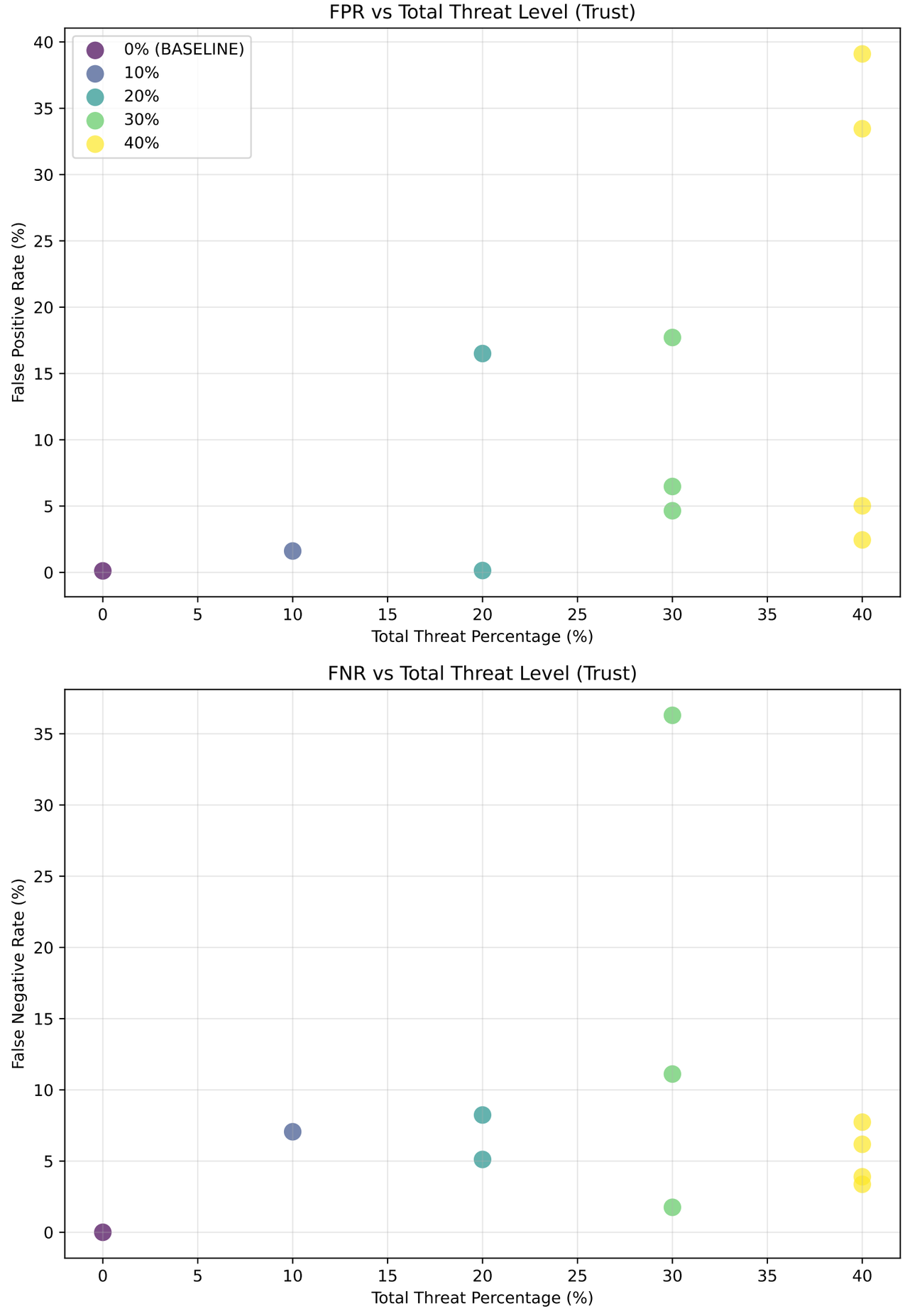
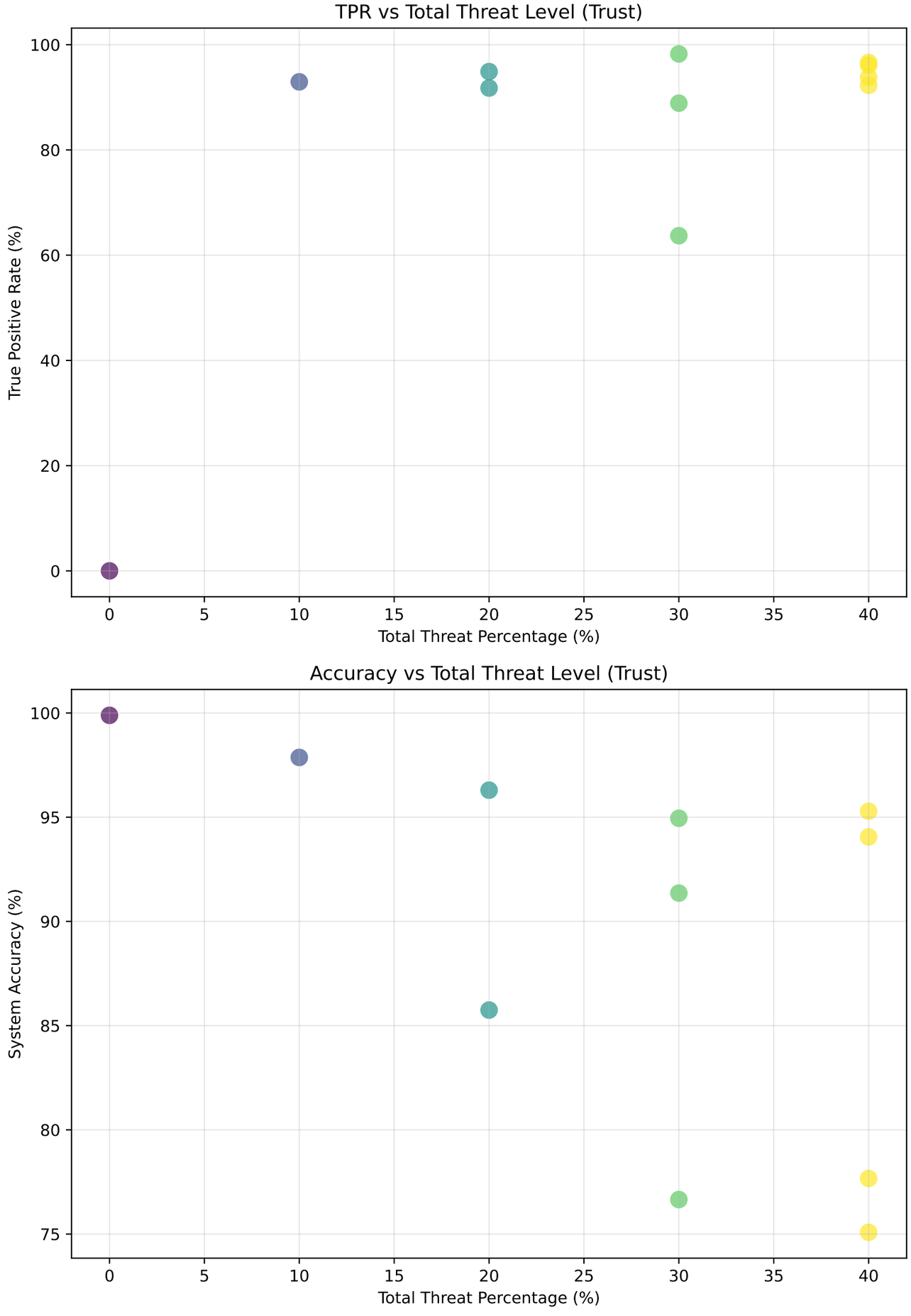
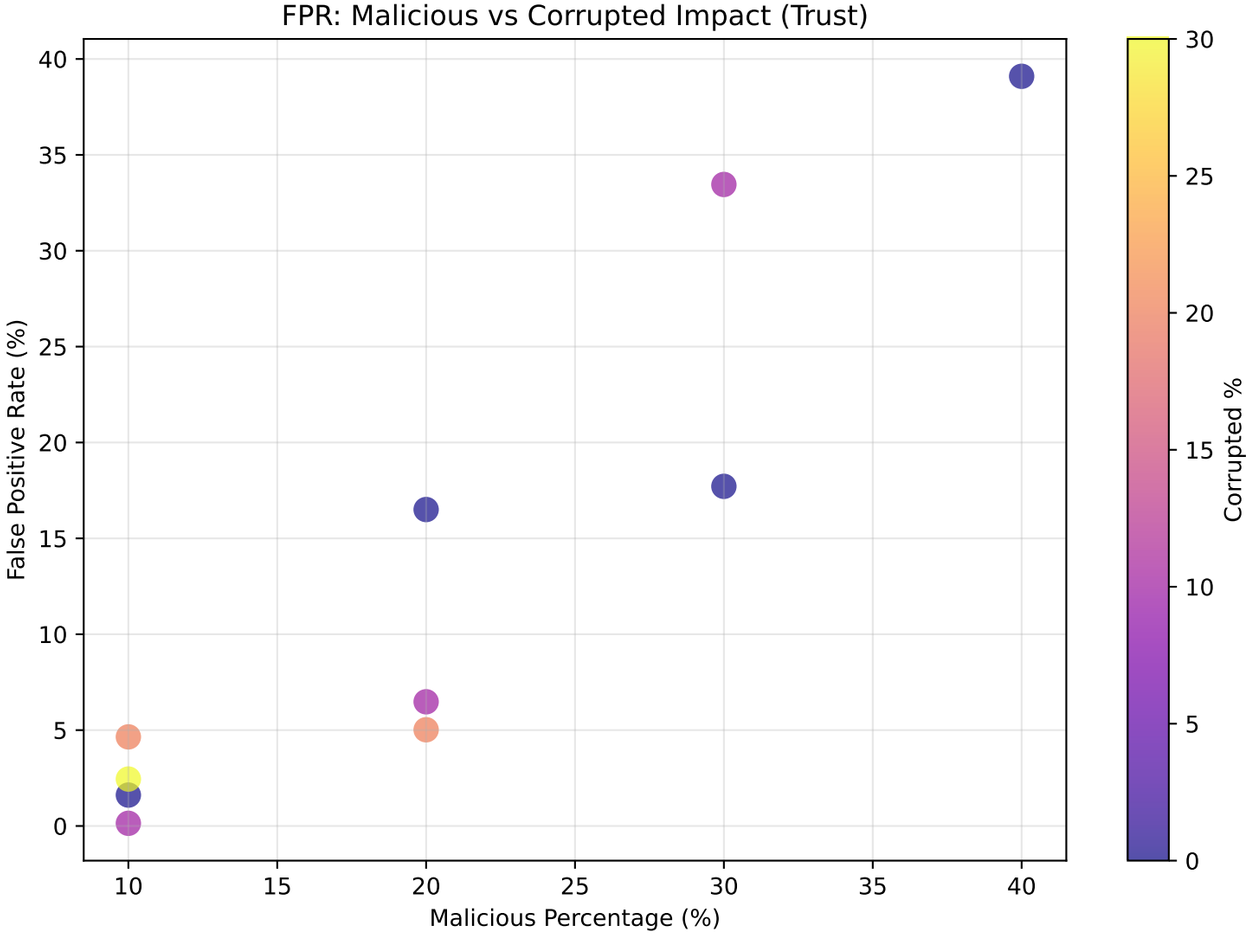
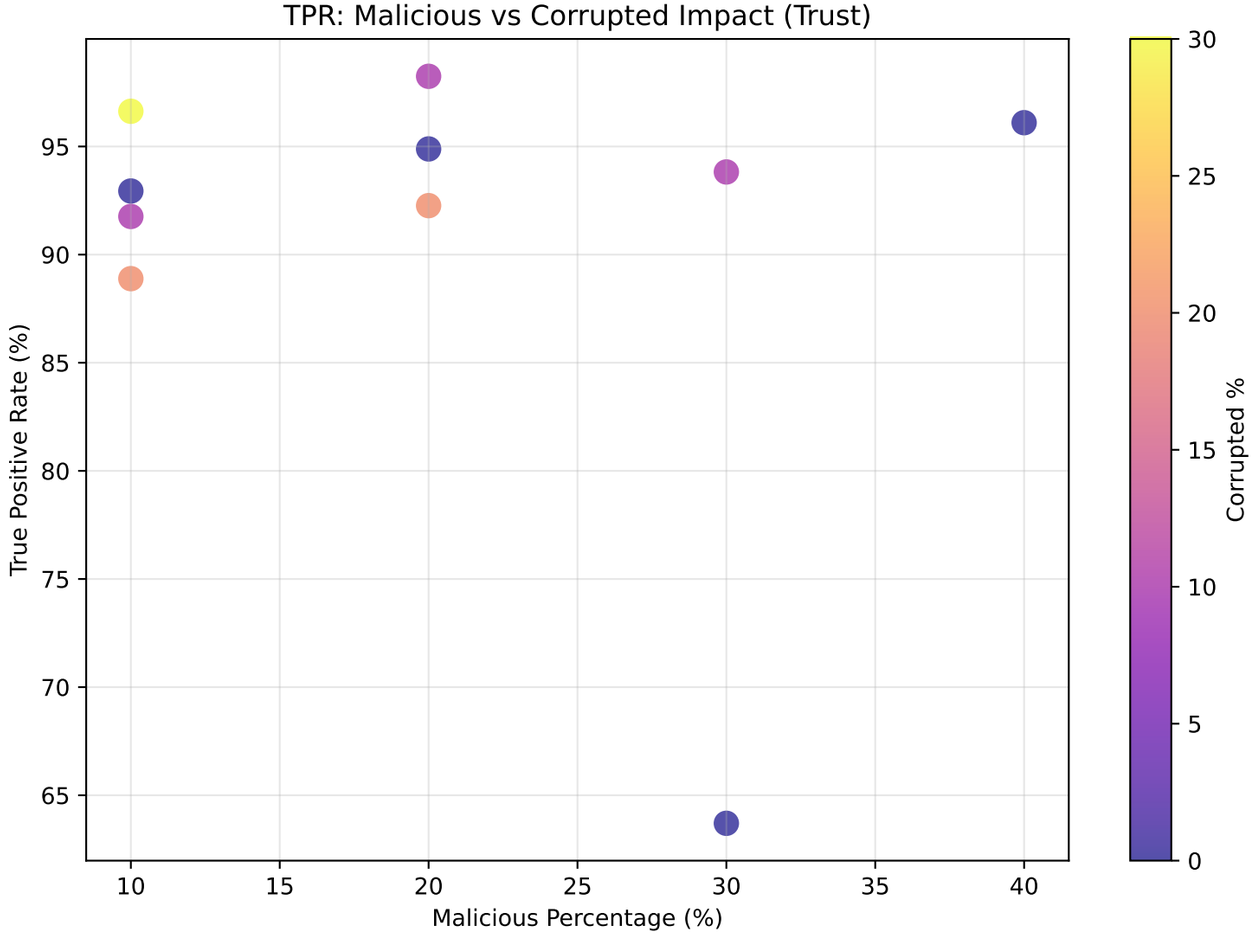
Trust Metrics vs Total Threat Level

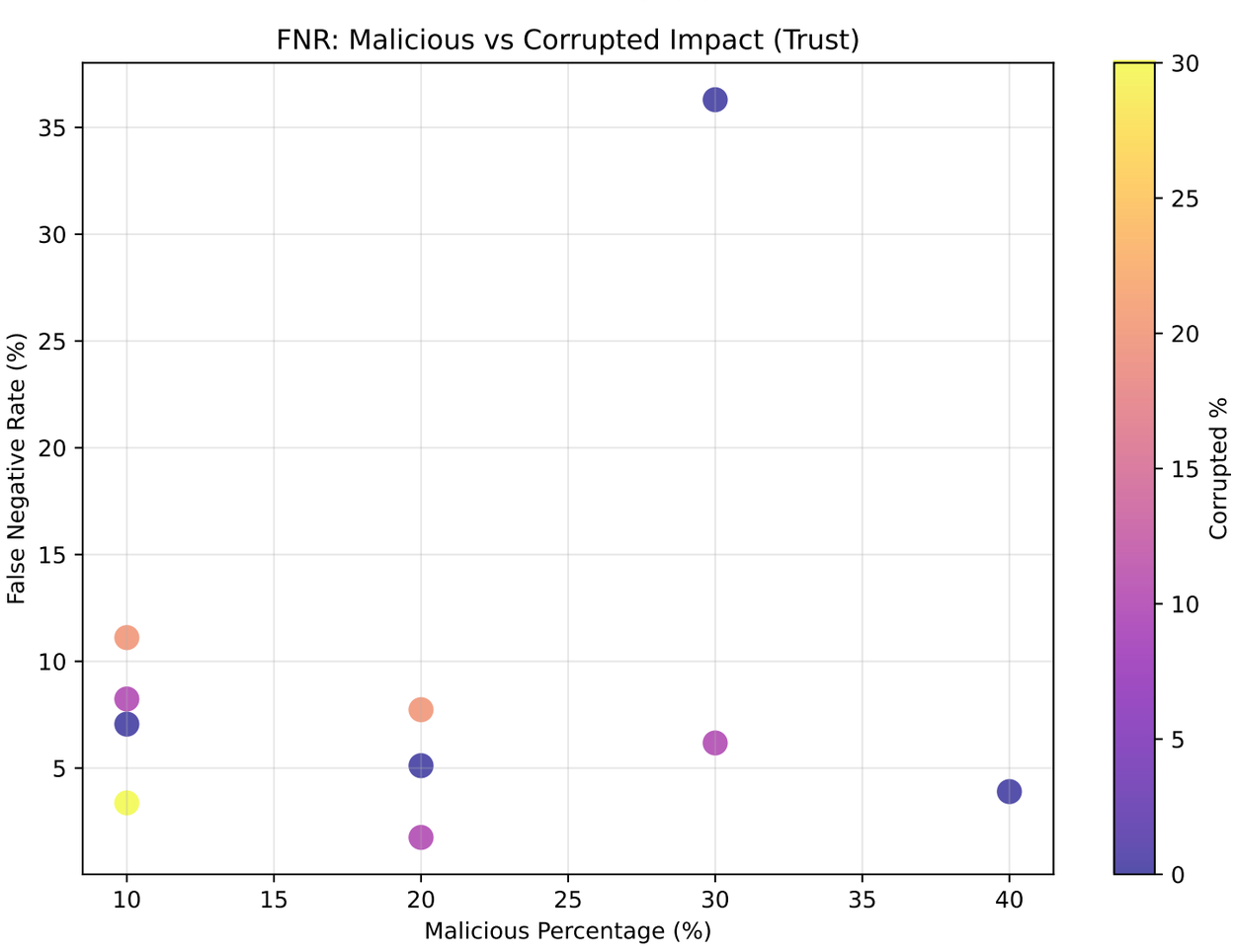


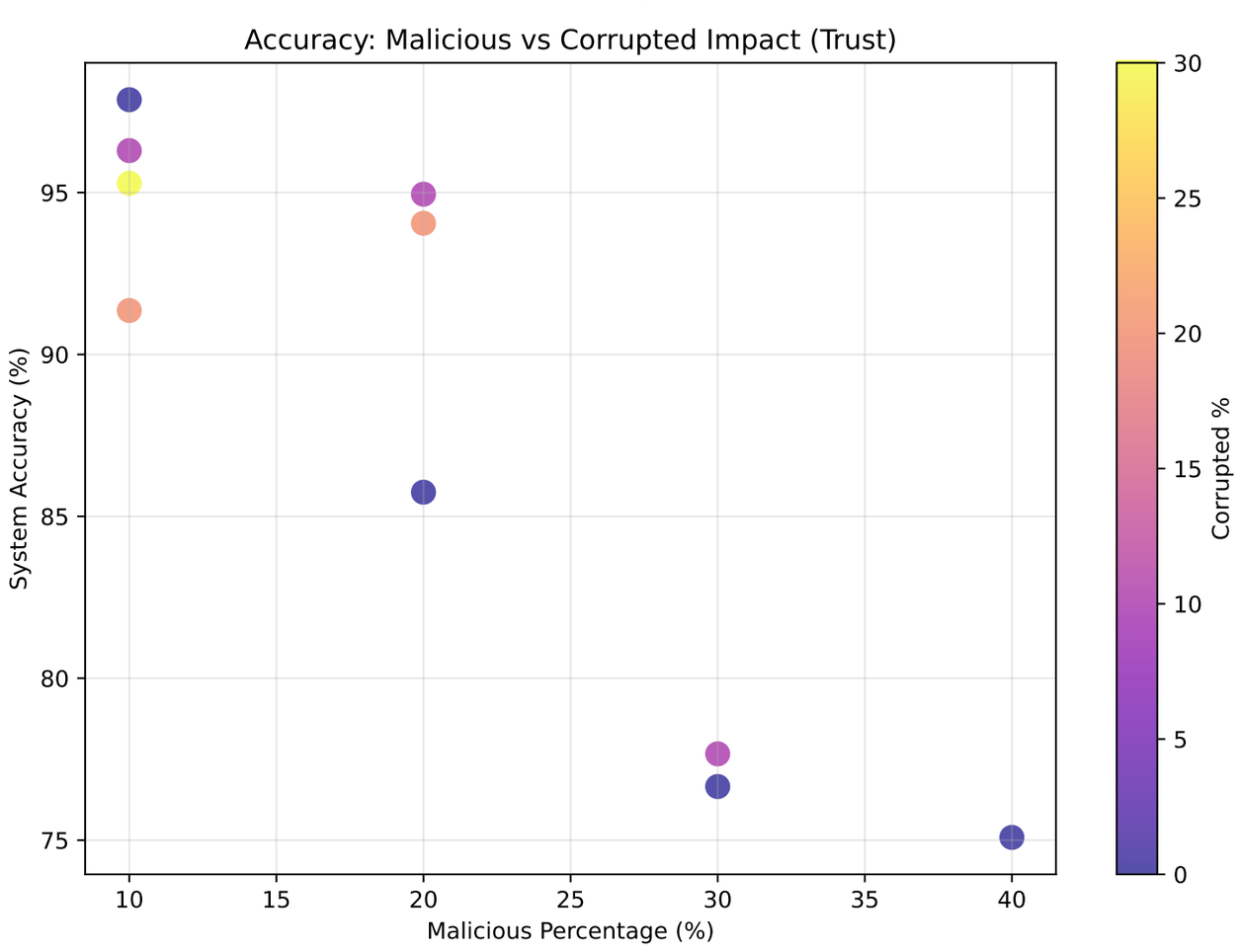


Trust Analysis: Malicious vs Corrupted Percentage Impact

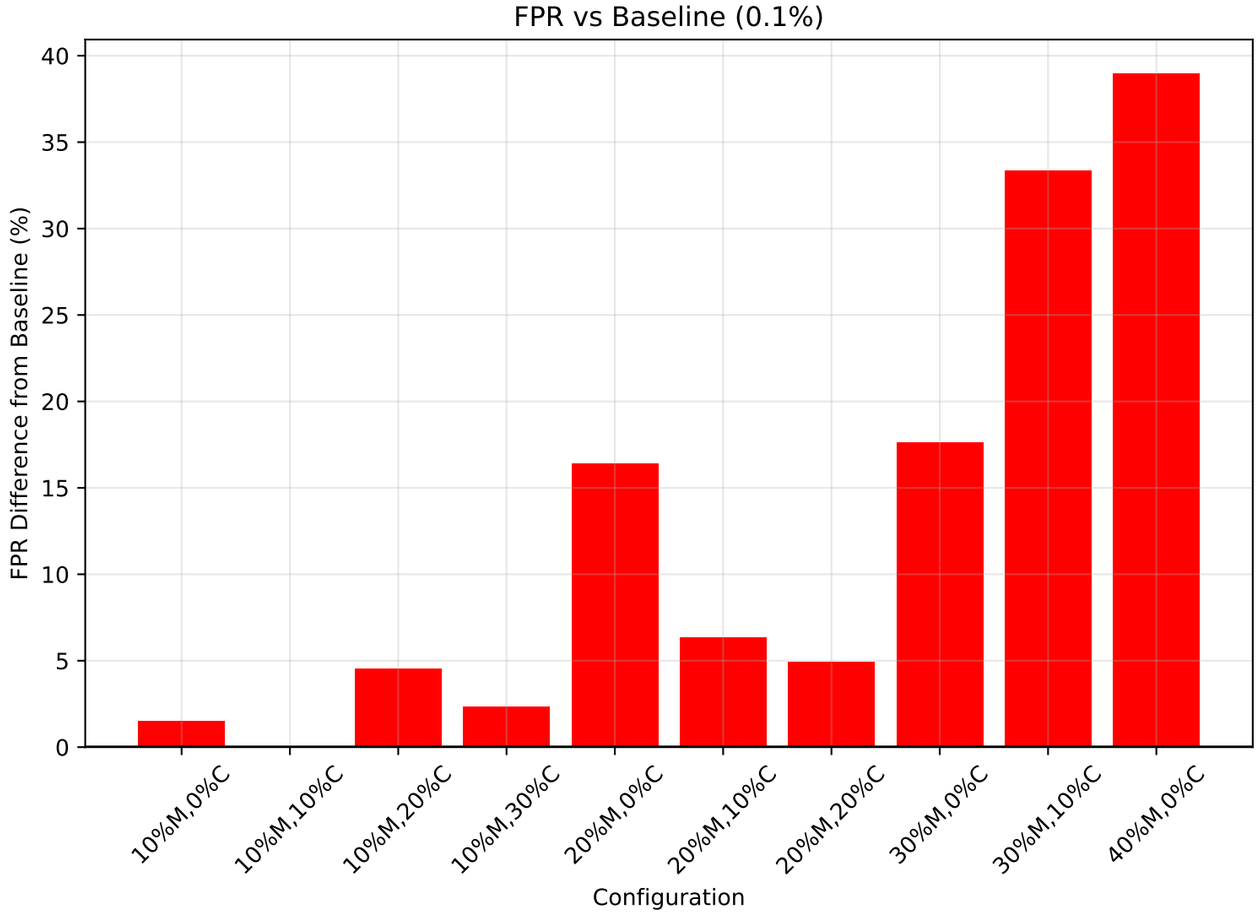


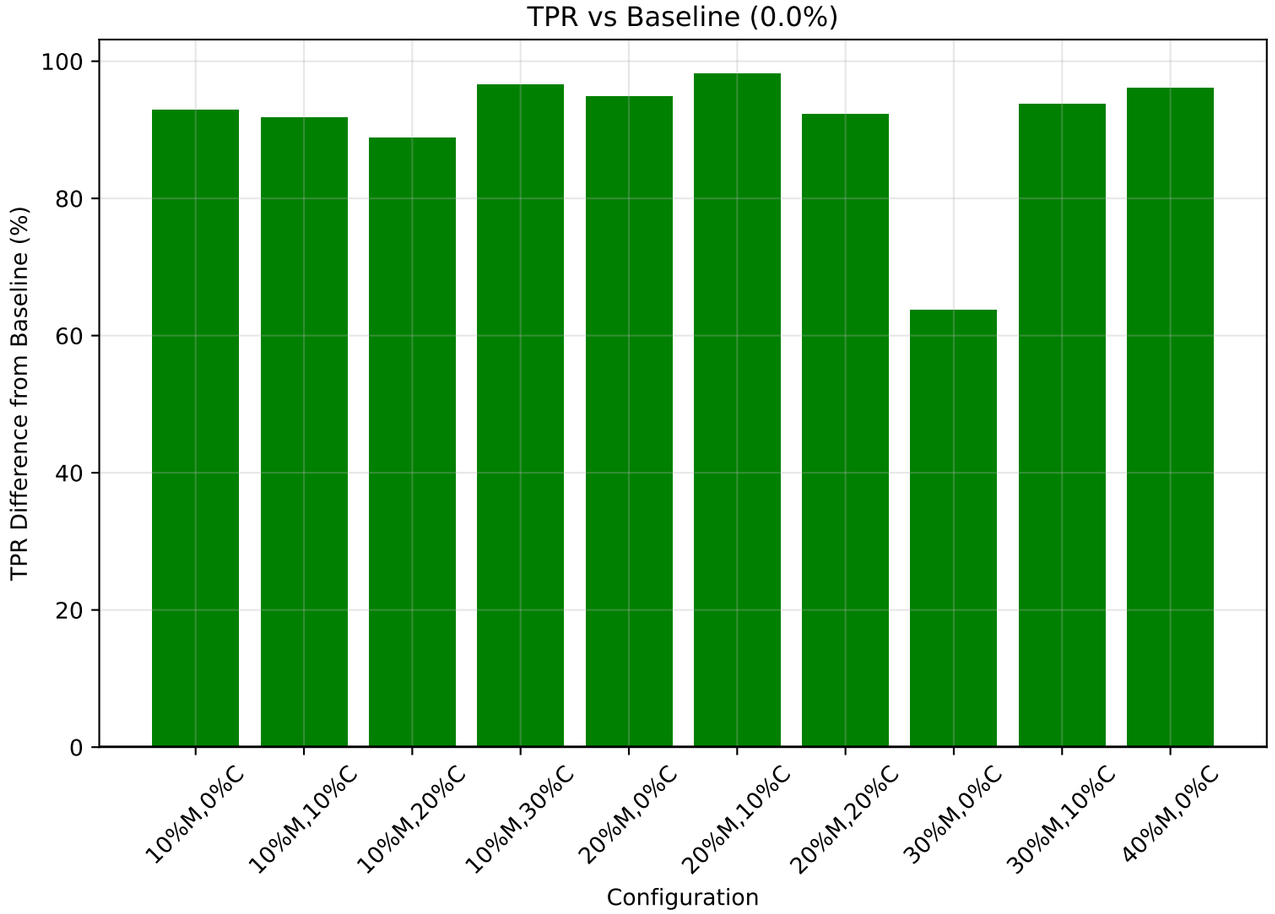


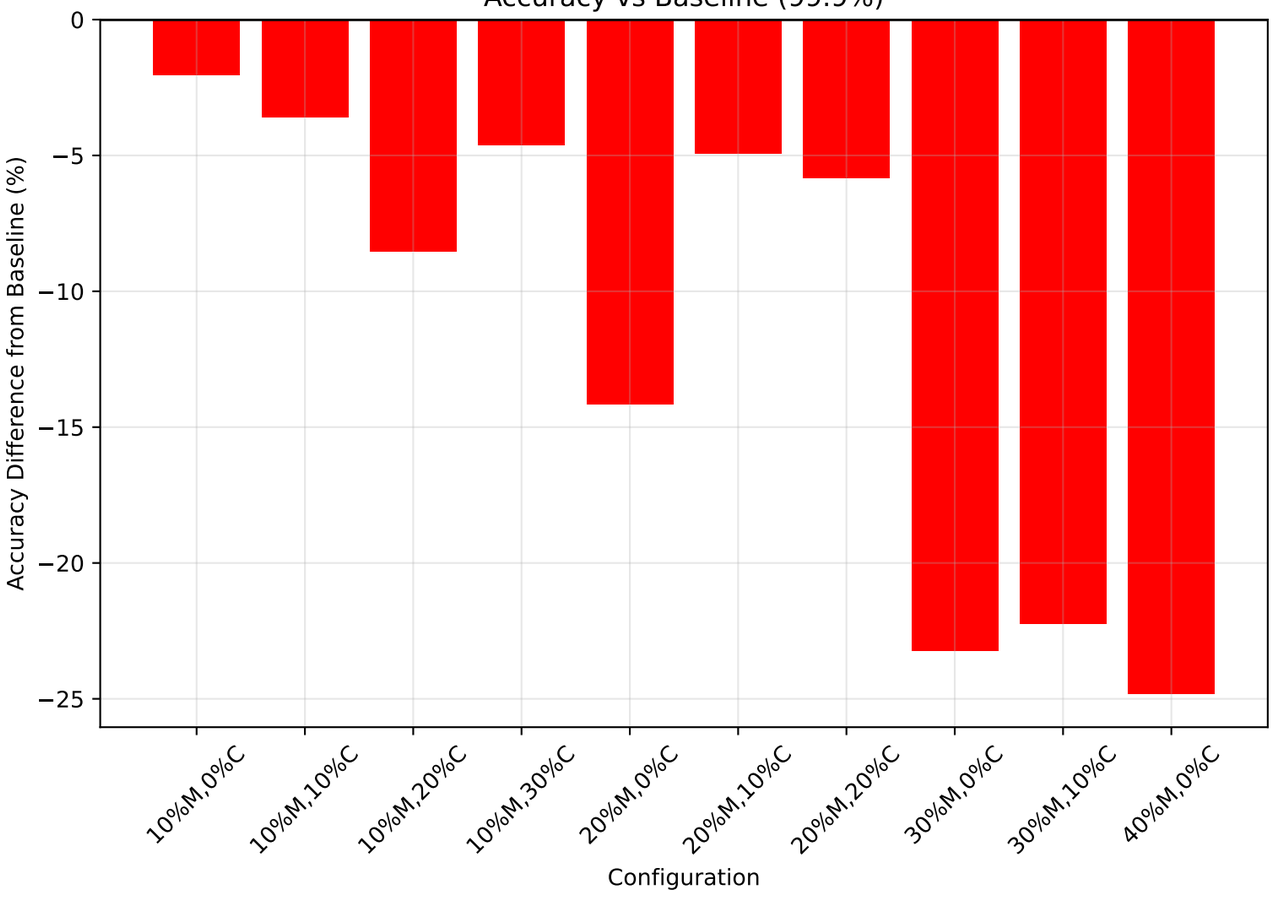


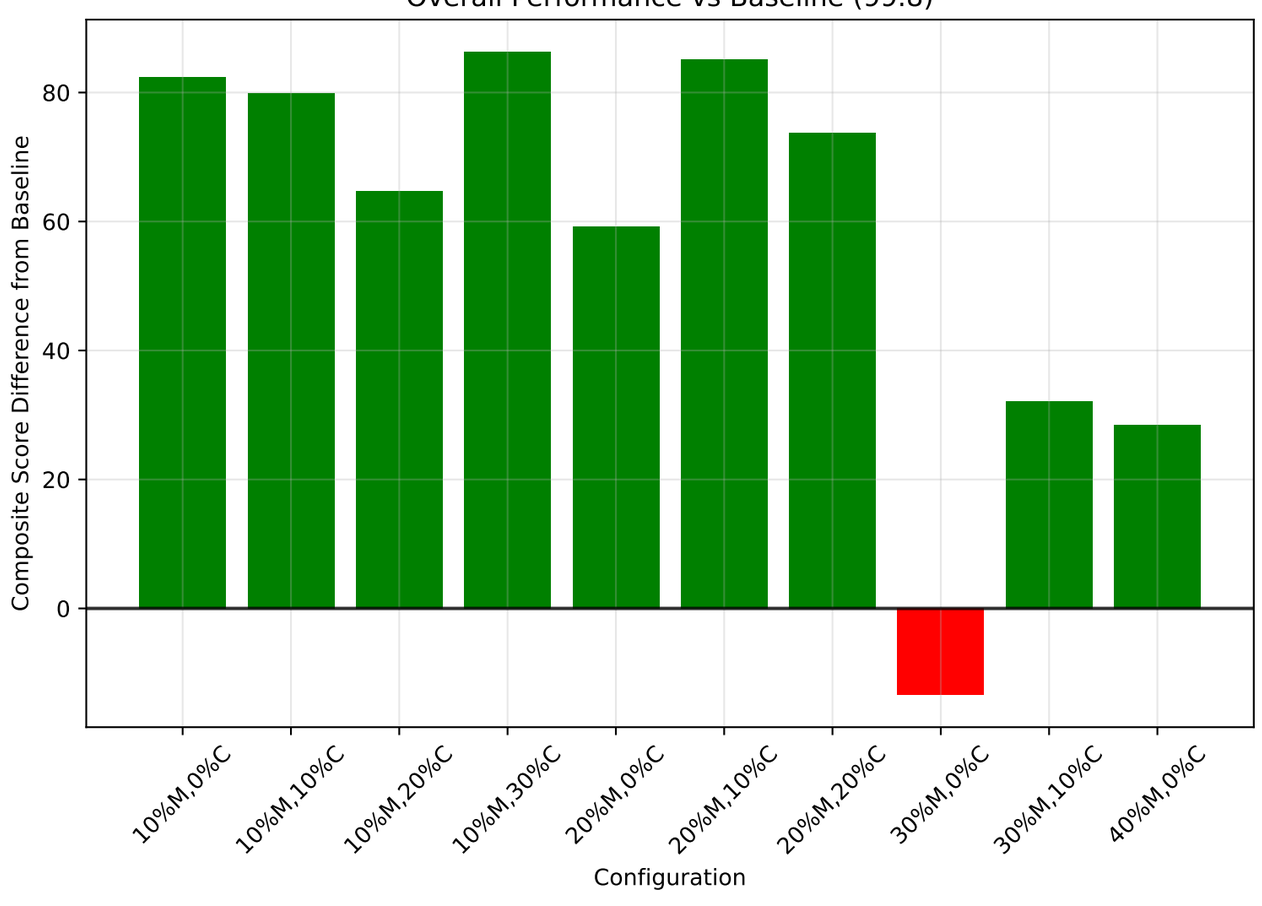


Trust Performance vs Baseline Analysis

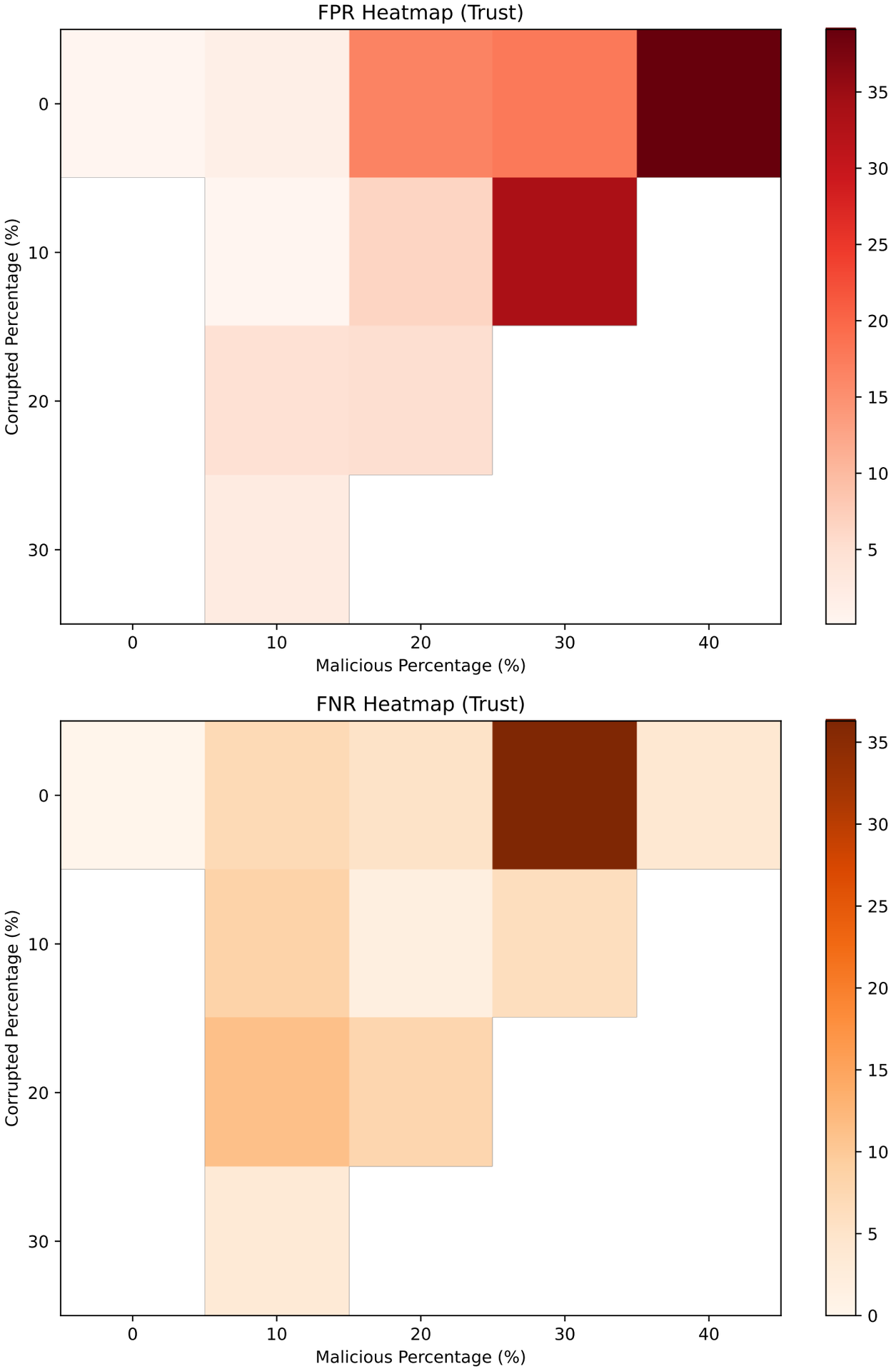


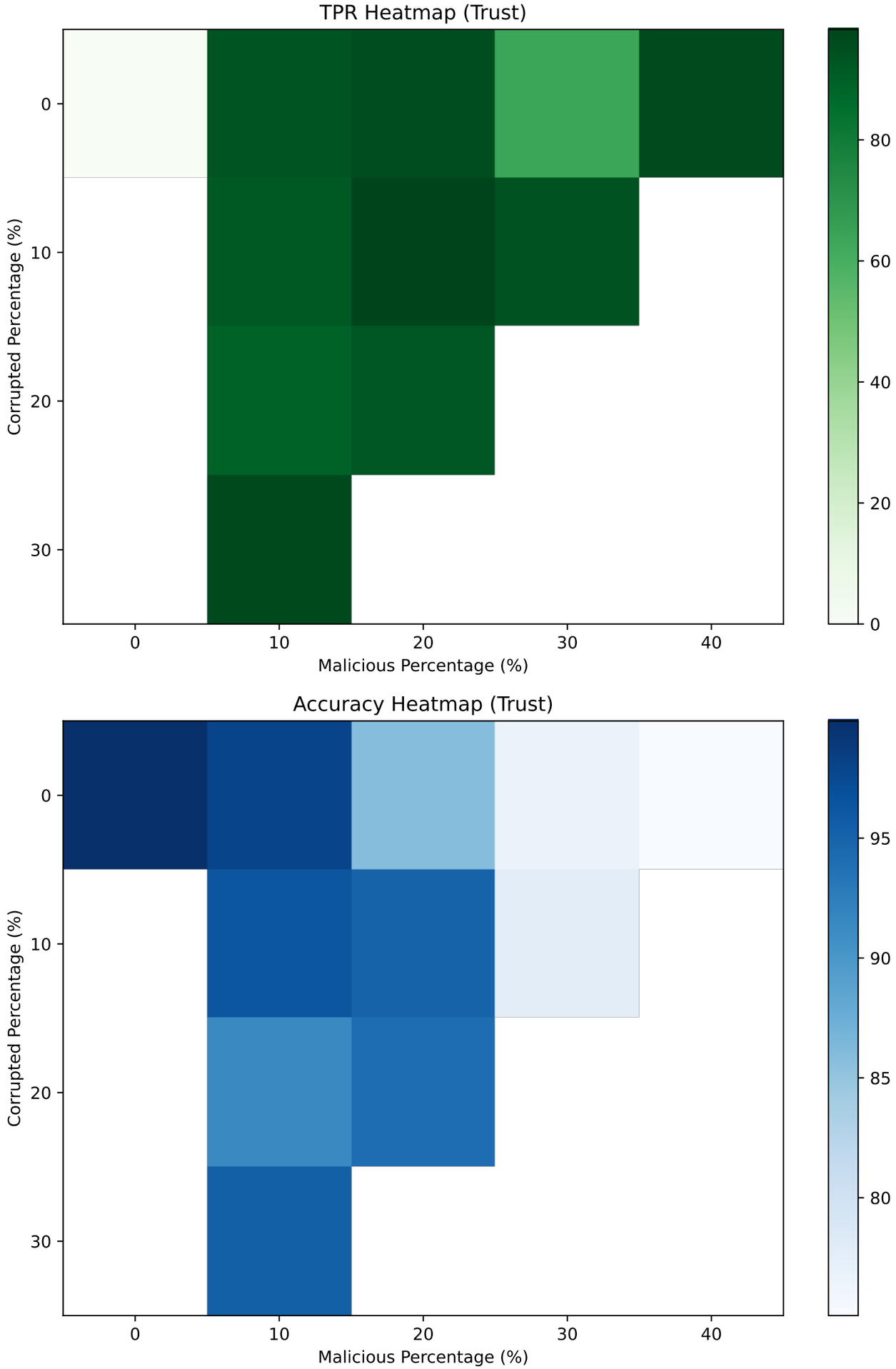






Trust Performance Heatmaps





Trust Configuration Effectiveness Ranking

```
TRUST PERCENTAGE CONFIGURATION RANKING
==============================================

1. 10%M, 30%C (Total: 40%)
   FPR: 2.4%, TPR: 96.6%, FNR: 3.4%, Acc: 95.3%
   Score: 186.1

2. 20%M, 10%C (Total: 30%)
   FPR: 6.5%, TPR: 98.2%, FNR: 1.8%, Acc: 94.9%
   Score: 185.0

3. 10%M, 0%C (Total: 10%)
   FPR: 1.6%, TPR: 92.9%, FNR: 7.1%, Acc: 97.9%
   Score: 182.1

4. 10%M, 10%C (Total: 20%)
   FPR: 0.1%, TPR: 91.8%, FNR: 8.2%, Acc: 96.3%
   Score: 179.7

5. 20%M, 20%C (Total: 40%)
   FPR: 5.0%, TPR: 92.3%, FNR: 7.7%, Acc: 94.1%
   Score: 173.6

6. 10%M, 20%C (Total: 30%)
   FPR: 4.6%, TPR: 88.9%, FNR: 11.1%, Acc: 91.4%
   Score: 164.5

7. 20%M, 0%C (Total: 20%)
   FPR: 16.5%, TPR: 94.9%, FNR: 5.1%, Acc: 85.7%
   Score: 159.0

8. 30%M, 10%C (Total: 40%)
   FPR: 33.5%, TPR: 93.8%, FNR: 6.2%, Acc: 77.7%
   Score: 131.9

9. 40%M, 0%C (Total: 40%)
   FPR: 39.1%, TPR: 96.1%, FNR: 3.9%, Acc: 75.1%
   Score: 128.2

10. 0%M, 0%C (BASELINE)
   FPR: 0.1%, TPR: 0.0%, FNR: 0.0%, Acc: 99.9%
   Score: 99.8

11. 30%M, 0%C (Total: 30%)
   FPR: 17.7%, TPR: 63.7%, FNR: 36.3%, Acc: 76.7%
   Score: 86.3


TRUST PERCENTAGE TESTING:
This analysis tests different combinations of malicious
and corrupted peer percentages including a baseline
case (0% threats) to understand trust degradation.

CONFIGURATIONS TESTED:
Total configurations: 11
Includes baseline (0% threats) and threat levels 10-40%
```

Trust Summary Statistics

```
TRUST PERCENTAGE COMPARISON ANALYSIS - SUMMARY
==================================================

Generated: 2025-09-02 15:07:44
Results Path: FL/results_percentages_trust

CONFIGURATIONS ANALYZED:
• Total configurations: 11
• Malicious percentages: [np.int64(0), np.int64(10), np.int64(20), np.int64(30), np.int64(40)]
• Corrupted percentages: [np.int64(0), np.int64(10), np.int64(20), np.int64(30)]
• Total threat levels: [np.int64(0), np.int64(10), np.int64(20), np.int64(30), np.int64(40)]
• Includes baseline: True

DETAILED RESULTS:
\n• 0%M, 0%C (BASELINE):\n  FPR=0.1%, TPR=0.0%, FNR=0.0%, Acc=99.9%\n\n• 10%M, 0%C (Total: 10%):\n  FPR=1.6%, TPR=92.9%, FNR=7.1%, Acc=97.9%\n\n• 10%M, 10%C (Total: 20%):\n  FPR=8.1%, TPR=91.8%, FNR=8.2%, Acc=96.3%\n\n• 20%M, 0%C (Total: 20%):\n  FPR=16.5%, TPR=94.9%, FNR=5.1%, Acc=85.7%\n\n• 10%M, 20%C (Total: 30%):\n  FPR=4.6%, TPR=88.9%, FNR=11.1%, Acc=91.4%\n\n• 20%M, 10%C (Total: 30%):\n  FPR=6.5%, TPR=98.2%, FNR=1.8%, Acc=94.9%\n\n• 30%M, 0%C (Total: 30%):\n  FPR=17.7%, TPR=63.7%, FNR=36.3%, Acc=76.7%\n\n• 10%M, 30%C (Total: 40%):\n  FPR=2.4%, TPR=96.6%, FNR=3.4%, Acc=95.3%\n\n• 20%M, 20%C (Total: 40%):\n  FPR=5.0%, TPR=92.3%, FNR=7.7%, Acc=94.1%\n\n• 30%M, 10%C (Total: 40%):\n  FPR=33.5%, TPR=93.8%, FNR=6.2%, Acc=77.7%\n\n• 40%M, 0%C (Total: 40%):\n  FPR=39.1%, TPR=96.1%, FNR=3.9%, Acc=75.1%\n

TRUST METRICS EXPLANATION:
• FPR (False Positive Rate): Good nodes incorrectly removed / Total good nodes sent
• TPR (True Positive Rate): Malicious nodes correctly detected / Total malicious nodes sent
• FNR (False Negative Rate): Malicious nodes incorrectly confirmed / Total malicious nodes sent
• Accuracy: Overall system accuracy (correctly handled nodes / total nodes)

TRUST ANALYSIS INSIGHTS:
• Baseline case provides reference for system performance without threats
• Total threat level ranges from 0% (baseline) to 40%
• Trade-offs between malicious/corrupted at each threat level
• System performance degradation under varying adversarial conditions
• Trust evaluation under different threat scenarios

TRUST PERCENTAGE PARAMETERS:
The malicious_percentage and corrupted_percentage parameters control
the proportion of adversarial peers in the federated learning system:
• Malicious peers: Send bad updates to degrade model performance
• Corrupted peers: Have degraded data quality but not malicious intent
• Baseline: 0% threats - ideal trust scenario
• Total threat = malicious% + corrupted%
• Higher percentages create more challenging trust environments
```